PLAYGROUND DESIGN PROPOSAL

DARESALAM PRIMARY AND MIDLEVEL SCHOOL

SITE PARAMETERS

School	2 of 6
Total Students	85
Compound Area	78,243 m²
KG Boundary	638 m²
Buildings (KG)	0 (adjacent)
Open Space (KG)	≈ 638 m²

EXISTING EQUIPMENT

- 1 Slide — Functional (needs paint)

EQUIPMENT GAPS

- 1 Merry-Go-Round — Nonfunctional (Remove)
- See-Saw — Missing
- Swing — Missing
- Balance Beam — Missing
- Climbing Structure — Missing

Site analysis, concept design, and detailed design proposal for pre-primary playground infrastructure serving 85 children aged 5 to 6 years within a 638 m² dedicated pre-primary compound. The KG boundary is the primary design area, situated within a larger 78,243 m² school campus. This medium-small school follows the Minimum Viable / Phased approach (Option B) to optimise resource allocation.

Prepared for: Cities Alliance & Assosa City Education Office
Implementing Partner: International Rescue Committee (IRC)

DOCUMENT REF: SUIDAC-PD-002
REVISION 1.0 · JULY 2026

PHASE 1 — SITE ANALYSIS

1.1 Site Context Report

Daresalam Primary and Midlevel School is situated within Assosa City, the capital of Benishangul-Gumuz Regional State in western Ethiopia. The school compound occupies a large **78,243 m² parcel**, with multiple service and residential zones within the boundary. The pre-primary section serves **85 students** (40 male, 45 female) aged approximately 5 to 6 years, classifying it as a medium-small school within the SUIDAC programme.

The critical design parameter for this site is the **638 m² KG boundary**, a dedicated pre-primary playground area that is the primary focus of this design. Unlike the larger school compounds where buildings are distributed across the site, the KG boundary at Daresalam contains **zero building footprint** — all school buildings are located in the adjacent main school area, outside the KG boundary polygon. This means the entire 638 m² is theoretically available for playground infrastructure, though setbacks from surrounding buildings and the boundary fence must be maintained.

The compound boundary includes 39 line features delineating multiple internal zones, including service areas, residential quarters, and designated future development zones. From VLM image analysis, 5–6 permanent structures are identified in the northwest corner of the compound, with buildings occupying approximately 40 percent of the total land area. A large central open play space exists within the main school grounds, and a Phase 2 future zone is identified at the southern point of the compound. The KG boundary polygon is positioned adjacent to the main school buildings, providing close proximity for teacher supervision from the pre-primary classrooms.

The existing play equipment inventory within the KG boundary is minimal: only **one functional slide** that requires repainting. No swings, see-saws, balance beams, or climbing structures are present. A **nonfunctional merry-go-round** has been assessed as structurally unsafe and will be removed prior to any new construction. Five critical equipment categories are absent: swing, see-saw, balance beam, climbing structure, and tunnel crawl. The tunnel crawl and climbing structure are deferred to Phase 2 under the phased implementation approach.

Climate conditions at Assosa are characterised by a mean annual temperature of approximately 22 °C, with daily maxima occasionally exceeding 35 °C during the long dry season (October through May). Annual rainfall averages 1,200 mm, concentrated in the kiremt rainy season from June to September. These conditions necessitate robust shade provision within all play zones and a gravity-based drainage system to prevent standing water accumulation, which poses both safety and health risks for young children.

1.2 Constraint Map Description

The site has been divided into four colour-coded constraint zones based on the overlay analysis of the KML boundary data (KG boundary polygon and school compound boundary with 39 line features), VLM image interpretation, and field observations. These zones guide all subsequent layout decisions and equipment placement within the 638 m² KG boundary.

Zone	Colour	Description	Area (m²)	Constraint
RED — No-Go	Red	Fence setback zone (1.5 m from KG boundary fence line), building wall setbacks (3.0 m from adjacent school buildings). Includes nonfunctional merry-go-round removal zone. Swing and slide platforms require 3.0 m clearance if placed near boundary edge.	≈ 95	Hard exclusion

Zone	Colour	Description	Area (m^2)	Constraint
YELLOW — Caution	Amber	Transition zones adjacent to building walls from the main school area (3.0 m buffer). Ground near fence line requiring grading. Areas where existing slide is located requiring retrofit inspection.	≈ 120	Conditional use
GREEN — Available	Green	Central open area within KG boundary, clear of buildings and setback zones. Primary targets for new equipment installation and phased expansion. Approximately 60% of KG boundary.	≈ 383	Preferred
BLUE — Inf rastructure	Blue	Pathway corridors (1.2 m min width), drainage swales, and utility connection points. Entry/exit transition zone connecting KG boundary to main school compound.	≈ 40	Service reserve

Table 1. Constraint zone classification and area allocation within KG boundary.

1.3 Opportunity Assessment

Daresalam School presents a favourable opportunity profile for playground development. While the total compound is very large (78,243 m^2), the design focus is the 638 m^2 KG boundary — a compact but adequate area for the 85-student pre-primary cohort. The active play area requirement is 80 m^2 per 50 students, scaled proportionally: 80 × (85/50) = **136 m^2 minimum**. With an additional 20 percent quiet/shade zone allocation of approximately **34 m^2**, the total minimum play infrastructure area is approximately **170 m^2**. This is well within the 638 m^2 KG boundary, leaving substantial room for generous equipment spacing, wide safety zones, and clearly marked Phase 2 expansion areas.

The zero-building-footprint condition within the KG boundary is a significant advantage. Unlike other school sites where buildings create complex setback geometries and visual obstructions, the entire KG boundary at Daresalam is open terrain. This simplifies layout planning, maximises usable equipment surface area, and ensures unobstructed teacher sight-lines from the adjacent pre-primary classrooms. The adjacency to the main school buildings provides natural supervision capability — teachers can monitor the playground from classroom doorways without leaving the building.

The large school compound surrounding the KG boundary provides additional practical benefits: material storage areas during construction, space for community labour staging, and proximity to existing school infrastructure (water supply, sanitation). The 39 internal line features within the compound indicate an organised spatial hierarchy with defined circulation corridors, service zones, and residential areas, suggesting good site management capacity within the school administration.

The single existing slide, while functional, requires anti-rust treatment (zinc-chromate primer and enamel topcoat) and repainting. Its retention significantly reduces Phase 1 procurement costs. The removal of the nonfunctional merry-go-round will free approximately 16 m^2 of space for new equipment installation. The compact KG boundary means all equipment placement must be carefully coordinated to avoid safety zone overlap while maintaining the 3.0 m clearance from the adjacent school buildings.

PHASE 2 — CONCEPT DESIGN

2.1 Design Criteria and Scaling

All concept designs have been developed in accordance with the non-negotiable design standards specified in the IRC SUIDAC Playground Design Manual. Equipment dimensions, safety zone clearances, surface treatment depths, and material specifications follow the prescribed values exactly. The following scaling calculations apply to Daresalam Primary and Midlevel School's pre-primary enrolment of 85 students:

- **Active Play Zone:** 80 m² per 50 students × (85/50) = **136 m² minimum**
- **Quiet/Shade Zone:** 20% of total play area = 0.20 × 170 ≈ **34 m²**
- **Total Play Infrastructure Area:** ≈ **170 m²** (within 638 m² KG boundary)
- **Pathways:** Estimated 30 m × 1.2 m = **36 m²**
- **Minimum Slides Required:** 1 per 50 students × (85/50) = **1.7 → 2 slides** (1 existing + 1 new in Phase 2)
- **Minimum Swings Required:** 1 per 30 students × (85/30) = **2.83 → 3 seats** (1 new double bay)

> Material Treatment Requirements: All metal components shall receive zinc-chromate primer followed by enamel topcoat (2 coats). All timber and bamboo shall undergo boron-based anti-termite treatment via 24-hour soak before installation. River sand surfacing shall be washed and graded to 150–300 mm depth under all equipment. Pathways shall use fine gravel at 100 mm compacted depth.

2.2 Option A — Maximum Play Value (Reference Only)

Option A represents the theoretical maximum play value configuration for a school of this size, providing the full complement of equipment types specified in the SUIDAC manual. This option includes two slides, three swing seats across two bays, one merry-go-round (replacement), two see-saws, two climbing structures, two tunnel crawls, two balance beams, and stepping logs — totalling 16 individual installations across 8 equipment categories. While this option provides the most comprehensive play experience, it represents an over-investment for a school serving 85 children within a constrained 638 m² KG boundary.

The principal disadvantage of Option A for Daresalam is the equipment density relative to the KG boundary size. With a total equipment surface area of approximately 195 m² under Option A, the active play zone would consume over 30 percent of the total KG boundary area, leaving insufficient space for adequate circulation, quiet/shade zones, pathways, and Phase 2 expansion. The merry-go-round replacement alone requires a 4.3 m diameter safety zone (16 m²), consuming space disproportionate to its play value. For these reasons, Option A is presented here as a reference for future Phase 2 expansion but is **not recommended as the initial implementation strategy**.

ID	Equipment	Qty	Status	Surface Area (m²)
E1	Merry-Go-Round	1	New replacement	16.8
E2	See-Saw	2	New	44.0
E3	Slide	2	1 existing + 1 new	21.0
E4	Chain Swing	2 bays (3 seats)	New	35.3
L1	Balance Beam	2	New	15.7

ID	Equipment	Qty	Status	Surface Area (m²)
L2	Climbing Structure	2	New (bamboo dome)	19.2
L3	Tunnel Crawl	2	New	13.5
L4	Stepping Logs	1 set	New	11.5

Table 2. Option A equipment schedule — Maximum Play Value (reference only).

2.3 Option B — Minimum Viable / Phased (SELECTED)

Given the medium-small student population of 85 children and the constrained 638 m² KG boundary, **Option B: Minimum Viable / Phased** has been selected as the recommended design for Daresalam Primary and Midlevel School. This option provides the essential core equipment needed to deliver a comprehensive developmental play experience while optimising resource allocation. Phase 1 retains the one functional existing slide (E3) and adds three critical missing pieces: one new swing (E4), one new see-saw (E2), and one new balance beam (L1). The nonfunctional merry-go-round is removed. Phase 2 additions include one climbing structure (L2), one tunnel crawl (L3), and one merry-go-round (E1, replacement).

The total Phase 1 equipment surface area is approximately **82 m²** of washed river sand, which, combined with interstitial circulation space, provides an effective active play area of approximately **136 m²** — meeting the scaled minimum requirement of 136 m². This compact layout occupies approximately 21 percent of the KG boundary, leaving ample room for a 34 m² quiet/shade zone, pathways, and a clearly demarcated Phase 2 expansion area. The layout is concentrated in the central portion of the KG boundary, maximising distance from the fence setback zones and adjacent school buildings.

The pathway network under Option B consists of a single 1.5 m wide main path connecting the pre-primary classroom entrance to the active play zone, with a secondary 1.2 m path connecting the active zone to the quiet/shade area. Drainage follows the same gravity-based principle as all SUIDAC designs, with a shallow swale along the main path alignment directing surface water toward the fence line collection point. The total estimated cost of Phase 1 is approximately 35 percent of a full Option A implementation, with Phase 2 expansion costing an additional 55 percent of the Phase 1 budget when funding becomes available.

Item	Phase 1 (Core)	Phase 2 (Expansion)	Total
Slides	1 (existing retained)	1 new	2
Chain Swings	1 new (2 seats)	1 new bay (1 seat)	3 seats
See-Saws	1 new	1 new	2
Climbing Structure	—	1 new (bamboo)	1
Merry-Go-Round	Remove (nonfunctional)	1 new (replacement)	1
Balance Beams	1 new	1 new	2
Tunnel Crawl	—	1 new	1
Quiet/Shade Zone	34 m²	10 m² expansion	44 m²

Table 3. Phased equipment plan for Option B — Minimum Viable (Selected).

2.3.1 Phase 1 Surface Area Calculations

Each equipment piece requires a defined surface area including the equipment footprint plus the mandatory safety zone. The following calculations detail the Phase 1 surface requirements:

Equipment	Footprint (m²)	Safety Zone (m²)	Total Surface (m²)	Surface Type
E3 Slide (×1, existing)	3.0 × 1.5 = 4.5	6.0	10.5	River sand
E4 Chain Swing (×1, new)	6.25	12.5	18.8 (≈ 19.0)	River sand
E2 See-Saw (×1, new)	4.0 × 0.5 = 2.0	8.0 × 3.0 − 2.0 = 22.0	24.0	River sand
L1 Balance Beam (×1, new)	3.5 × 0.25 = 0.9	3.5 × 2.0 = 7.0	7.9 (≈ 8.0)	River sand
E1 Merry-Go-Round (removal)	—	—	0 (removed)	N/A

Table 4. Phase 1 surface area calculations per equipment type (Option B).

Total Phase 1 equipment surface area: 61.5 m² (rounded to 62 m²) of washed river sand at 150–300 mm average depth. Combined with approximately 74 m² of interstitial circulation space, the effective active play area reaches approximately 136 m², satisfying the scaled minimum requirement of 136 m².

2.3.2 Safety Zone Verification

All safety zones have been verified against the mandatory minimum clearances. The following checks confirm compliance with the IRC SUIDAC Playground Design Manual:

Check Item	Standard	Design Value	Status
Fall zone around moving equipment	≥ 2.0 m	2.0 m (swing, see-saw)	PASS
Clearance from fence line	≥ 1.5 m	1.5 m maintained from KG boundary	PASS
Clearance: swings/slides to buildings	≥ 3.0 m	≥ 3.0 m from adjacent school bldgs	PASS
Pathway width	≥ 1.2 m	1.5 m main, 1.2 m secondary	PASS
Equipment height limit	≤ 2.5 m	Max 2.0 m (slide)	PASS
Safety zone overlap (inter-equipment)	None permitted	All verified clear	PASS

Table 5. Safety zone compliance verification matrix.

2.3.3 Supervision Sight-Line Analysis

Teacher supervision is a critical safety requirement for pre-primary playgrounds. The zero-building condition within the KG boundary is a significant advantage — no internal structures obstruct sight-lines. All Phase 1 equipment is arranged within the central portion of the 638 m² boundary, ensuring that the maximum sight-line distance from the adjacent pre-primary classroom to any equipment piece is approximately **20 metres** — well within the recommended 30 metre maximum for unobstructed visual supervision.

The swing (E4), which presents the highest supervision challenge due to its dynamic movement range, is positioned closest to the pre-primary classroom at approximately 8 m distance. The existing slide (E3) is positioned at an intermediate distance of 12–15 m, while the see-saw (E2) and balance beam (L1) are placed at approximately 18–20 m from the building but remain fully visible due to the absence of any visual obstructions within the flat, open KG boundary. A single teacher stationed at the classroom doorway can observe all children on all equipment simultaneously without any blind spots.

2.3.4 Drainage Management

The drainage strategy employs a minimum 1.0 percent slope across all play zones within the KG boundary, graded from north (near adjacent school buildings) to south (toward fence line). Surface water flows via shallow swales (150 mm deep, 300 mm wide) along the pathway network toward a collection point at the southern fence line. From there, water discharges through a grated outlet into the existing compound drainage. The compact Option B layout reduces the total swale length to approximately 20 m, lowering both construction effort and long-term maintenance requirements.

Under each equipment installation, the 150–300 mm layer of washed river sand serves as both a fall-attenuation surface and a percolation medium. The sand layer is underlain by geotextile fabric that prevents mixing with the native lateritic subsoil while allowing water to drain through freely. This dual-function approach eliminates the need for sub-surface drainage pipes and can be maintained by community labour through periodic sand loosening and top-up every 6–12 months.

PHASE 3 — DETAILED DESIGN (OPTION B SELECTED)

3.1 Master Layout Plan Description

The master layout for Daresalam Primary and Midlevel School organises the playground into four distinct functional zones within the 638 m² KG boundary. The layout prioritises teacher sight-lines, natural drainage, and compact spatial efficiency appropriate for an 85-child cohort, while maintaining the required 3.0 m clearance from the adjacent school buildings and 1.5 m setback from the KG boundary fence.

Zone 1 — Transition Area (Northern Perimeter): A 3.0 m wide transition strip extends along the northern edge of the KG boundary, adjacent to the main school buildings. This strip provides dry, safe access between the pre-primary classrooms and the play areas. It is surfaced with compacted fine gravel (100 mm depth) over geotextile fabric. No equipment is installed within this zone to maintain the mandatory building setback clearance.

Zone 2 — Active Play Area (Central): The primary play zone occupies approximately 136 m² (including circulation) in the central portion of the KG boundary. Equipment is arranged in a compact semi-circular pattern with the swing (E4) positioned closest to the pre-primary classroom for maximum supervision proximity. The existing slide (E3) is positioned on the eastern flank, relocated slightly to optimise orientation and surface drainage. The new see-saw (E2) is placed on the western side of the active zone, oriented along the east-west axis. The new balance beam (L1) is positioned at the southern edge of the active zone. A 1.5 m wide main pathway bisects the zone from north to south, providing both circulation and emergency access.

Zone 3 — Quiet/Shade Zone (Southern): A 34 m² quiet zone is designated in the southern portion of the KG boundary, positioned away from the dynamic equipment. Simple log seating and a shade structure are installed for reading, rest, and low-activity play. Where boundary trees exist along the fence line, these provide natural canopy cover. Additional shade may be provided through a simple timber frame with shade cloth. This zone is separated from the active play area by a 1.2 m gravel pathway that serves as a physical and visual buffer.

Zone 4 — Phase 2 Expansion Area (Eastern): A clearly demarcated expansion zone of approximately 120 m² is reserved east of the active play area for future Phase 2 installations (climbing structure, tunnel crawl, and merry-go-round replacement). This zone is marked with low timber bollards (400 mm height, 500 mm spacing) and surfaced with compacted fine gravel at 80 mm depth. In the interim period, it serves as informal open play space for running games and group activities. No permanent installations are permitted in this zone until Phase 2 funding is confirmed.

3.2 Equipment Placement Schedule

The following table provides relative placement coordinates for each Phase 1 equipment installation. Coordinates are referenced from the southwestern corner of the KG boundary fence line (origin: 0, 0), with X increasing eastward and Y increasing northward. All positions maintain the required safety clearances from the adjacent school buildings, KG boundary fence, and adjacent equipment.

ID	Equipment	Qty	Zone	Rel. Position (X, Y) m	Orientation	Notes
E3	Slide (existing)	1	2	(16, 12)	S-facing	Refurbished, anti-rust + repaint
E4	Chain Swing (new)	1	2	(8, 14)	N/A	New double-bay, 2 seats

ID	Equipment	Qty	Zone	Rel. Position (X, Y) m	Orientation	Notes
E2	See-Saw (new)	1	2	(18, 8)	E-W axis	New installation, 4.0 m length
L1	Balance Beam (new)	1	2	(12, 5)	N-S axis	New, 3.5 m length

Table 6. Phase 1 equipment placement schedule with relative coordinates (Option B).

3.3 Grading and Earthwork Notes

The existing ground surface within the KG boundary is natural earth/soil typical of the Assosa region, generally flat but requiring grading to achieve the minimum 1.0 percent slope for drainage and to create level equipment platforms. The total earthwork volume is modest due to the compact Option B footprint.

Earthwork Item	Description	Estimated Volume	Method
Site clearing	Removal of overgrown vegetation, debris from KG boundary play zone and pathways	N/A (labour)	Hand tools, wheelbarrow
Hazard removal	Removal of nonfunctional merry-go-round from KG boundary	N/A (labour)	Bolt cutters, hand winch
Slope correction	Grade play zone to 1.0% fall from north to south within KG boundary	≈ 12 m^3 cut/fill	Hand-compacted laterite
Equipment platforms	Level pads under each equipment installation, 100 mm above surrounding grade for drainage	≈ 4 m^3 fill	Compacted gravel base
Drainage swale	150 mm deep × 300 mm wide swale along southern edge, 20 m length	≈ 0.9 m^3 excavation	Hand-dug, geotextile lined
Pathway sub-base	Fine gravel, 100 mm compacted over geotextile, 30 m total length	≈ 3.6 m^3	Imported gravel, hand-compacted
Sand surfacing	Washed river sand, 200 mm avg depth under all equipment zones (62 m^2)	≈ 12.4 m^3	Imported, levelled by hand
Tree protection mounds	Circular earth mounds around any preserved boundary tree trunks within KG boundary	≈ 0.5 m^3	Hand-shaped

Table 7. Grading and earthwork schedule.

Total estimated earthwork volume: 33 m^3 (cut/fill). All earthwork shall be executed by community labour using hand tools (shovels, rakes, hand tampers). No mechanical equipment is required. The compact Option B footprint significantly reduces earthwork compared to a full Option A implementation. Cut-and-fill balance is achieved by reusing excavated swale material as fill for slope correction and equipment platforms.

A compaction standard of 95 percent Modified Proctor Density shall be achieved for all filled areas using hand tampers. Quality assurance shall be performed by the site supervisor using a simple probe test (steel rod penetration resistance) at 3 m intervals across all graded areas.

3.4 Planting Plan

The planting strategy focuses on three objectives: supplementary shade provision within the quiet zone, erosion control on graded surfaces, and visual enhancement of the playground environment. All species are selected for local availability within 50 km of Assosa, drought tolerance, non-toxicity to children, and low maintenance requirements.

Species (Local Name)	Type	Qty	Location	Purpose	Spacing
Ficus sycomorus (Shola)	Deciduous shade tree	2	Southern boundary, within quiet zone of KG boundary	Supplementary shade canopy	8 m
Terminalia brownii (Dhidhessa)	Semi-deciduous tree	1	Eastern expansion zone, adjacent to drainage swale	Root stabilisation for swale banks	—
Codiaeum variegatum (local ornamental)	Low shrub	6	Along pathway edges, alternating sides	Visual definition, windbreak	1.5 m
Cynodon dactylon (Bermuda grass)	Ground cover	60 m²	Quiet zone and expansion areas within KG boundary	Erosion control, dust suppression	Broadcast seed
Euphorbia tirucalli (local hedge)	Low hedge (0.4 m)	12 m length	Boundary between active and quiet zones	Visual separation, no climbing risk	0.5 m (linear)

Table 8. Planting schedule — locally sourced species.

Any existing boundary trees along the KG fence line shall be preserved in their entirety. No pruning shall be performed without arborist assessment. Root protection zones of 1.0 m radius around each trunk shall be maintained during grading and construction. Circular earth mounds shall be built around the trunks to prevent soil compaction and root damage from foot traffic. Where possible, shade structures for the quiet zone may be supplemented with simple timber frames and shade cloth if natural canopy coverage is insufficient.

3.5 Maintenance Framework

Long-term playground sustainability depends on a clear maintenance protocol that can be executed by school staff and community volunteers with basic tools. The compact Option B layout reduces the total maintenance burden. The following framework defines maintenance tasks, frequencies, responsible parties, and material requirements.

Task	Frequency	Responsible	Materials/Tools	Notes
Sand loosening and top-up	Monthly	School groundskeeper	Rake, wheelbarrow, river sand	Check depth, replace if below 150 mm
Metal equipment rust check	Quarterly	School maintenance lead	Wire brush, primer, enamel paint	Touch-up zinc-chromate + enamel coat
Timber/bamboo termite check	Bi-annually (pre/post rain)	Community volunteer team	Boron solution, brush	Re-soak if soft spots detected
Fastener and joint inspection	Monthly	School groundskeeper	Wrench set, replacement bolts	Check swing chains, slide anchors, see-saw pivot
Drainage swale clearing	After each rain event	School staff	Shovel, hand trowel	Remove debris, check geotextile integrity

Task	Frequency	Responsible	Materials/Tools	Notes
Pathway gravel replenishment	Bi-annually	Community volunteer team	Fine gravel, rake	Top up to 100 mm, re-compact
Tree watering (new plants)	Daily for first 3 months	Student water monitors (supervised)	Watering cans	Morning watering, 5 L per tree per day
Fence integrity check	Quarterly	School maintenance lead	Pliers, wire, replacement posts	Check KG boundary fence for gaps, rust
Grass cutting in quiet zone	Monthly (rainy season)	School groundskeeper	Sickle or machete	Maintain ground cover below 100 mm height

Table 9. Maintenance schedule and responsibility matrix.

3.6 Material Sourcing — Local Availability

All materials specified in this design are available within a 50 km radius of Assosa city. This local sourcing requirement supports the community-labour construction model and minimises procurement lead times and transportation costs. The large school compound provides adequate space for material staging during construction.

Material	Source Location	Distance from Site	Estimated Cost Rate
Washed river sand	Blue Nile riverbed, south of Assosa	25 km	ETB 800/m³
Fine gravel (pathways)	Assosa municipal quarry	8 km	ETB 650/m³
Bamboo (construction grade)	Mao-Komo district forests	40 km	ETB 150/pole
Timber (Eucalyptus posts)	Assosa plantation nursery	12 km	ETB 200/post
Metal pipe (see-saw, swing frame)	Assosa metalworks workshop	5 km	ETB 3,500/piece
Zinc-chromate primer	Assosa hardware shops	3 km	ETB 450/gallon
Enamel topcoat paint	Assosa hardware shops	3 km	ETB 550/gallon
Boron treatment chemical	IRC supply chain (Addis Ababa)	Air freight	Included in IRC logistics
Geotextile fabric	IRC supply chain	Air freight + 5 km	Included in IRC logistics
Chain and hardware (swing)	Assosa metalworks / Bahir Dar	5–180 km	ETB 2,000/set
Timber bollards (expansion zone)	Assosa plantation nursery	12 km	ETB 150/bollard
Grass seed (Bermuda)	Assosa agricultural supply	5 km	ETB 200/kg

Table 10. Local material sourcing schedule.

3.7 Removal and Recycling Plan

Daresalam School has one significant removal item that must be addressed before new construction begins within the KG boundary: the nonfunctional merry-go-round. The existing slide requires refurbishment but will be retained in situ. The following plan details the removal, disposal, or recycling approach.

Item	Condition	Action	Disposal / Recycling	Estimated Effort
Merry-go-round	Nonfunctional, structurally unsafe; assessed for removal during site protocol	Full removal; cut bolts, dismantle on site	Scrap metal sold to Assosa recycler; revenue offsets labour costs	4 labour-hours, bolt cutters + hand winch
Existing slide	Functional but requires repainting; surface corrosion on frame	Retain in situ; refurbish with anti-rust treatment	N/A — retained equipment	3 labour-hours, wire brush + primer + enamel
Overgrown vegetation (KG boundary)	Natural ground cover; some areas above 300 mm height	Slash, clear, and rake from play zone	Compost on-site or transport to municipal green waste	8 labour-hours, sickles, rakes

Table 11. Removal and recycling plan for existing hazards.

The total estimated removal effort is **15 labour-hours**, achievable by a team of four community labourers over two working days. The scrap metal from the merry-go-round generates revenue through sale to the Assosa metal recycler, partially offsetting labour costs. The existing slide refurbishment (3 hours) represents a cost-effective alternative to procurement of a new slide, saving an estimated ETB 5,000 while maintaining equipment continuity.

End of Design Document — Daresalam Primary and Midlevel School (School 2 of 6)

This document was prepared under the IRC SUIDAC programme for Cities Alliance donor review. All dimensions, calculations, and specifications are site-specific to Daresalam Primary and Midlevel School and should not be applied to other school sites without independent verification.